Low Temperature Thermoelectric Properties of Hot Pressed Composites of $CrSb_2$: Evidences for Phonon-drag Effect.

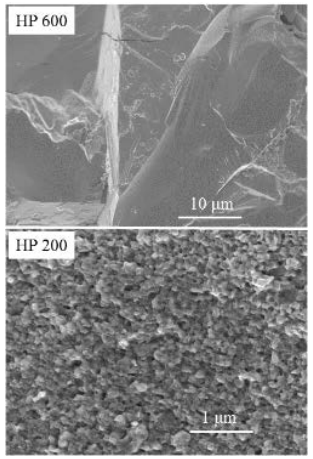
Mani Pokharel [1], Machhindra Koirala [2], Zhifeng Ren [3], and Cyril Opeil [4]

[1]Mercer University, Macon, GA 31206; [2]Rensselaer Polytechnic Institute, Troy, NY 12180; [3]University of Huston, Huston, TX 77204; [4]Boston College, chestnut Hill, MA 20467

Introduction: $CrSb_2$ has been one of the well studied compounds due to its interesting magnetic and electronic transport behavior. Recently, this strongly correlated narrow-gap semiconductor has attracted attention as a thermoelectric material for potential cooling application specifically because of its large Seebeck coefficient ($s \approx 4500\ \mu V/K$ at 18 K for single crystal samples). Despite the large Seebeck coefficient the ZT values for $CrSb_2$ single crystal are rather low due to very high thermal conductivity. While hot-pressing has been proven to be very effective technique to suppress the thermal conductivity of many thermoelectic materials, it also provides a tool to study the strength of correlation between the electronic and the phononic properties of the material. In this study we investigate the low temperature thermoelectric properties of samples of $CrSb_2$ prepared by hot-press densification at two chosen temperatures 200 and 600 °C. Our results show dramatically reduced thermal conductivities in hot-pressed samples. However, the peak value in Seebeck coefficient suffers a simultaneous suppression thereby indicating a striking interrelationship between the thermal conductivity and the Seebeck coefficient. This kind of simultaneous and proportionate change in thermal conductivity and Seebeck coefficient provide a strong evidence for a significant presence of phonon-drag effect in this system. Furthermore, analysis shows that several other scenarios exist in this compound manifesting the phenomenon of phonon-drag.

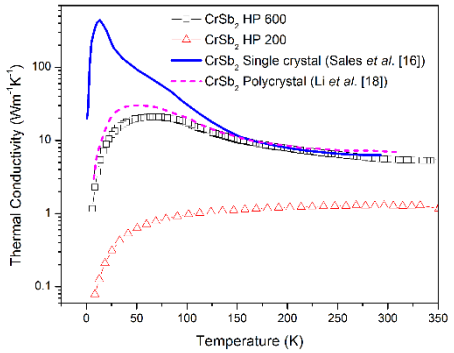
EXPERIMENTAL

- Resulting ingot obtained from solid-state reaction route was ball milled for 15 hours.
- Nanopowder was DC hot pressed for 2 minutes under a mechanical pressure of 19 Mpa.
- Thermoelectric properties were measured using thermal transport option (TTO) of PPMS.
- SEM images were taken of the fresh fracture of the disk pellet.
- Estimated average grain-size
  HP 600:10 μm
  HP 200:100 nm.



THERMAL CONDUCTIVITY

- $k \approx 0.78\ Wm^{-1}K^{-1}\ at\ 71\ K$ for HP 200, a reduction by 96% over the sample HP 600.
- $k \approx 0.13\ Wm^{-1}K^{-1}\ at$ 12 K(for HP 200), a reduction by three orders of magnitude over the reported large maximum of $k \approx 300 - 400\ Wm^{-1}K^{-1}\ near\ 12\ K$ (Sales et al.[1]) for single crystal sample.
- Kinetic theory: $k_{lattice} = \frac{1}{3} c_V \lambda_{phonon}\, v_s$ gives $l_{phonon} \approx 7\mu m\ at\ 18\ K$ [1].
- The electron mean free path determined within the Drude model: $l_{electron} = \frac{m^* v_e}{\rho n e^2}$ with $v_e = \sqrt{3k_B T/m^*}$ gives $l_{electron} \approx 8\ nm\ at\ 18\ K$ satisfying the condition $l_{electron} < grain\ size\ (d) < l_{phonon}$.
- Results are indicative of grain boundary scattering being the dominant mechanism responsible for thermal conductivity reduction in composite $CrSb_2$ samples.
- Note Casimir limit: $\tau^{-1}_{boundary} = \frac{v}{d}$



SEEBECK COEFFICIENT

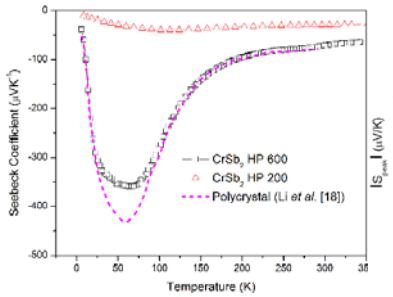
- Seebeck coefficient suffers a simultaneous and drastic reduction as the thermal conductivity decreases sharply.
- $s \approx -40\ \mu V/K$ for the sample HP 200 at $T_{peak} \approx 125\ K$.
- $|s_{max}|$ as a linear function of k_{peak} is a clear indication of direct inter-relationship between the Seebeck coefficient and the thermal conductivity.
- Noting that the thermal conductivity of $CrSb_2$ is lattice dominated with negligible contribution from the charge carriers, this scenario where the lattice thermal conductivity detects the Seebeck coefficient is, in general, the manifestation of the phonon-electron interaction also known as phonon-drag effect.
- In general ,

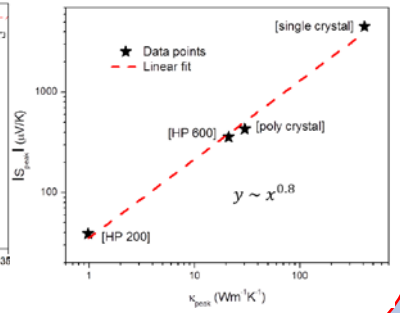
$$s = s_{diffusion} + s_{drag}$$

- For a semiconductor, in the first-order approximation,

$$s_{drag} \approx \frac{\beta\, \lambda_{phonon}\, v_s\, n\, e}{\sigma T} \quad \rightarrow \quad s_{drag} \propto k$$

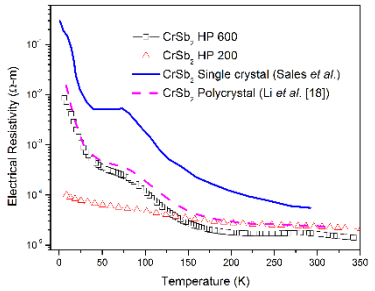
$$s_{diffusion} = \mp 86.2\left[ln\frac{4.7 \times 10^{15}}{n} + \frac{3}{2} ln\frac{m^*}{m} + \frac{|\Delta E|}{kT} + \frac{3}{2} lnT\right]$$

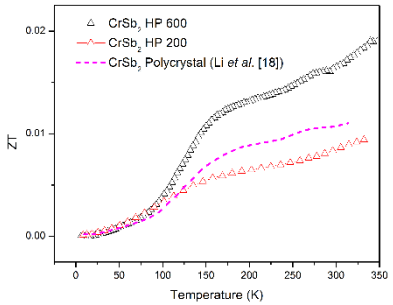




ELECTRICAL RESISTIVITY AND *ZT*

- At low temperature, electrical resistivity decreases in the nanocomposite samples.
- Thermal conductivity and the power factor (s^2/ρ) in $Z = s^2/\rho k$ contribute dynamically as a function of grain size.
- For the sample HP 600, the ZT values are improved. This improvement is due to moderately reduced thermal conductivity combined with moderately maintained power factor.





Ref: [1] B. C. sales, A. F. May, M. A. McGuire, M. B. Stone, D. J. Singh, and D. Mandrus, Phys. Rev. B, 86 (2012) 235136

Acknowledgement: We gratefully acknowledge funding for this work by the Department of Defense, United States Air Force Office of Scientific Research's MURI program under contract FA9550-10-1-0533.